

A Novel Scene-Matching Positioning Approach for Low-Altitude UAVs Considering Elevation Differences in Satellite Images

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Abstract—The precise localization of uncrewed aerial vehicles (UAVs) is a critical prerequisite for mission success in smart urban environments driven by the Internet of Things (IoT). Traditional Global Navigation Satellite System (GNSS)-based positioning schemes are vulnerable to signal interference and spoofing attacks, highlighting the importance of visual scene matching as a robust solution for anti-jamming localization. However, this technology faces challenges such as sensitivity to perspective variations and trajectory discontinuities. To address these limitations, this article presents a novel visual scene matching-based UAV localization framework. First, a spatiotemporally constrained candidate map retrieval strategy is proposed, which enables efficient and accurate candidate retrieval by coupling single-frame probability estimation with a hidden Markov model (HMM). Second, a pose recovery algorithm based on cross-frame elevation difference constraints is introduced, and by leveraging feature matches between adjacent UAV frames and between UAV frames and satellite images, the elevation information of satellite images is estimated and 2-D–3-D correspondences between UAV images and satellite images are established. This constructs a 3-D constraint model that enables the UAV to obtain 3-D coordinates independent of perspective variations, thereby maintaining localization accuracy under oblique views. Finally, a factor graph joint optimization framework is developed to fuse local relative poses with satellite-based absolute geographic poses, thereby ensuring smooth and stable localization performance. Extensive experiments on public and real-world datasets demonstrate that the proposed method significantly improves localization accuracy and validates its effectiveness in supporting collaborative IoT agent operations in GNSS-denied environments.

Index Terms—Elevation recovery, factor graph optimization, geolocalization, Global Navigation Satellite System (GNSS)-denied, satellite image, uncrewed aerial vehicles (UAVs).

I. INTRODUCTION

IN THE intelligent era driven by the Internet of Things (IoT), uncrewed aerial vehicles (UAVs) act as critical nodes

within the integrated air–ground–space IoT architecture, with applications in smart city management, emergency response, intelligent logistics, and other domains becoming increasingly prevalent. These IoT scenarios impose strict requirements on the real-time positioning accuracy and anti-jamming capabilities of UAVs. Traditional Global Navigation Satellite System (GNSS)-dependent positioning schemes often fail due to signal attenuation and spoofing attacks in GNSS-denied environments, including urban canyons surrounded by high-rise buildings and areas with strong electromagnetic interference, thereby severely compromising the mission reliability of UAVs in IoT applications [1], [2]. To overcome the bottleneck of GNSS dependency, visual geolocation technology—which aligns real-time imagery from onboard cameras with pre-stored satellite or UAV remote sensing (RS) images—has emerged as a primary solution for anti-jamming positioning in the IoT ecosystem. This technology enables autonomous positioning using only visual sensors, making it well-suited to the low-power and lightweight deployment requirements of IoT devices. The core techniques in this field involve homogeneous or heterogeneous image retrieval and matching, as well as subsequent position estimation. For homogeneous image-matching, generally speaking, the traditional feature points (such as SIFT [3], SURF [4], and ORB [5]) can achieve good results in most cases. However, due to the substantial differences in image-capturing time, sensor modalities, and lighting conditions between UAV and existing RS images, deep-learning-based feature extraction and matching methods (such as SuperPoint [6] and LightGlue [7]) are widely applied [8], [9], [10] to heterogeneous image matching. In addition, some researchers have proposed using global image descriptors to support image retrieval or scene recognition. A typical example is NetVLAD [11], which has been extensively studied and applied in recent years for visual localization research [12], [13], [14]. However, most existing approaches perform direct brute-force retrieval from satellite image databases, which makes them difficult to be used in real-time scenarios.

Regarding the process of position estimation, most existing methods rely exclusively on 2-D feature points extracted from satellite images, and these approaches assume a strictly nadir (downward-facing) UAV camera perspective [8], [10], [14], [15], [16]. However, maintaining a nadir view is often infeasible in real-world UAV operations due to complex environmental conditions and flight dynamics. When the UAV operates at oblique angles, the reliance on purely 2-D

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satellite features leads to a significant degradation in localization accuracy [17]. Although some studies [17], [18], [19] have introduced digital elevation models (DEMs) or digital surface models (DSMs) to reconstruct the elevation information in satellite imagery, the low spatial resolution of DEM results in substantial errors in urban environments, while the high-precision data acquisition of DSM is costly and poses challenges for adaptation in large-scale IoT deployment scenarios. Second, traditional single-frame positioning strategies are vulnerable to environmental texture changes, which often cause abrupt trajectory deviations and undermine the spatiotemporal consistency required for reliable IoT operations [8], [20].

To address the aforementioned challenges, this article proposes a novel UAV visual localization system tailored for IoT applications in low-altitude urban scenes, which achieves robust localization in GNSS-denied environments via three key innovations.

- 1) A spatiotemporally constrained candidate map retrieval strategy is proposed, enabling efficient and accurate retrieval of candidate maps by coupling single-frame probability estimation with a hidden Markov model (HMM). By leveraging this strategy, the proposed method can dynamically update the candidate satellite submaps from large-scale databases and improve retrieval efficiency compared with existing approaches.
- 2) A cross-frame elevation difference-constrained pose recovery algorithm is proposed, which leverages the visual overlap between adjacent UAV frames and satellite image feature matching chains to construct a 3-D pose constraint model, effectively addressing localization failures under oblique perspectives. The experimental results demonstrate that without relying on any external elevation model, the proposed method can reduce the localization error from 31.51 to 3.68 m even when the UAV camera is tilted to -70° .
- 3) A factor-graph-based joint optimization framework is introduced, which fuses simultaneous localization and mapping (SLAM)-based local relative poses with 3-D absolute geographic poses derived from our elevation-augmented 3-D satellite features, while a dynamically optimized Sim(3) transformation continuously suppresses redundant global scale drift. Extensive experiments demonstrate the effectiveness of this approach in maintaining accurate and consistent alignment. This integration effectively eliminates abrupt variations in single-frame estimations and yields smoother, more stable global localization trajectories.

The remainder of this article is organized as follows. Section II reviews related state-of-the-art technologies; Section III details the system architecture and core components of the proposed fusion algorithm; Section IV evaluates the performance of the proposed algorithm through experiments; and Section V summarizes the research findings and discusses future directions.

II. RELATED WORKS

In this section, we divide existing visual geolocalization methods into two main categories, those based on 2-D RS images and those that incorporate satellite image elevation. A critical review is provided to summarize the strengths and limitations of each category.

A. Visual Geolocalization Based on 2-D RS Images

Achieving visual geolocalization for UAVs in GNSS-denied environments requires establishing feature correspondences between UAV images and RS images. Early studies, such as [20] and [21], used traditional handcrafted features such as SIFT and HOG to perform UAV localization. Such approaches work when real-time captured images and existing images are obtained with the same visual sensors in most cases; however, due to the large domain gap between UAV and satellite images—caused by differences in sensor modalities, acquisition time, viewing angles, and illumination conditions—traditional features often fail to establish sufficient correspondences, limiting their practical applicability [1], [22]. In response, recent studies have leveraged deep-learning-based feature extractors and matchers, such as Superpoint [6], SuperGlue [23], and LightGlue [7], to obtain sufficient and accurate matching points in heterogeneous image matching scenarios. For example, wildnav [8] demonstrated a vision-based UAV geolocalization framework using learned local features to directly estimate geographic coordinates. Similarly, Wang et al. [10] introduced an image moment-based method for selecting aerial photographs suitable for geolocation, effectively filtering out images unsuitable for localization. These methods leverage deep-learning-based feature extraction and matching to establish local correspondences between UAV and satellite images, compute the homography matrix H , and directly derive the UAV's geographic coordinates from H .

However, the above methods typically assume that the initial position of the drone is known in advance, which significantly limits their scalability in real-world applications. To overcome this constraint, many recent studies [14], [24], [25], [26] have adopted coarse-to-fine strategies. For example, GLVL [15] uses a two-stage visual localization framework that first uses a large-scale retrieval module to identify satellite regions resembling the UAV's flight scene, followed by a fine-grained matching module to accurately estimate the UAV's coordinates, thereby achieving real-time and precise localization. Similarly, Qiu et al. [24] use the NetVLAD global descriptor [11] to retrieve visually similar satellite images for a given UAV image, and then apply SuperPoint and SuperGlue to extract fine-grained feature correspondences. These correspondences are used in conjunction with the geographic metadata embedded in the satellite images to localize the UAV. However, these methods rely on direct retrieval from a large satellite image database, resulting in unstable retrieval accuracy and are also prone to errors in locally self-similar scenarios.

In addition, SLAM, a widely adopted technique in robotic navigation and localization, has also been used in many UAV localization tasks [27], [28], [29], [30]. However, visual SLAM

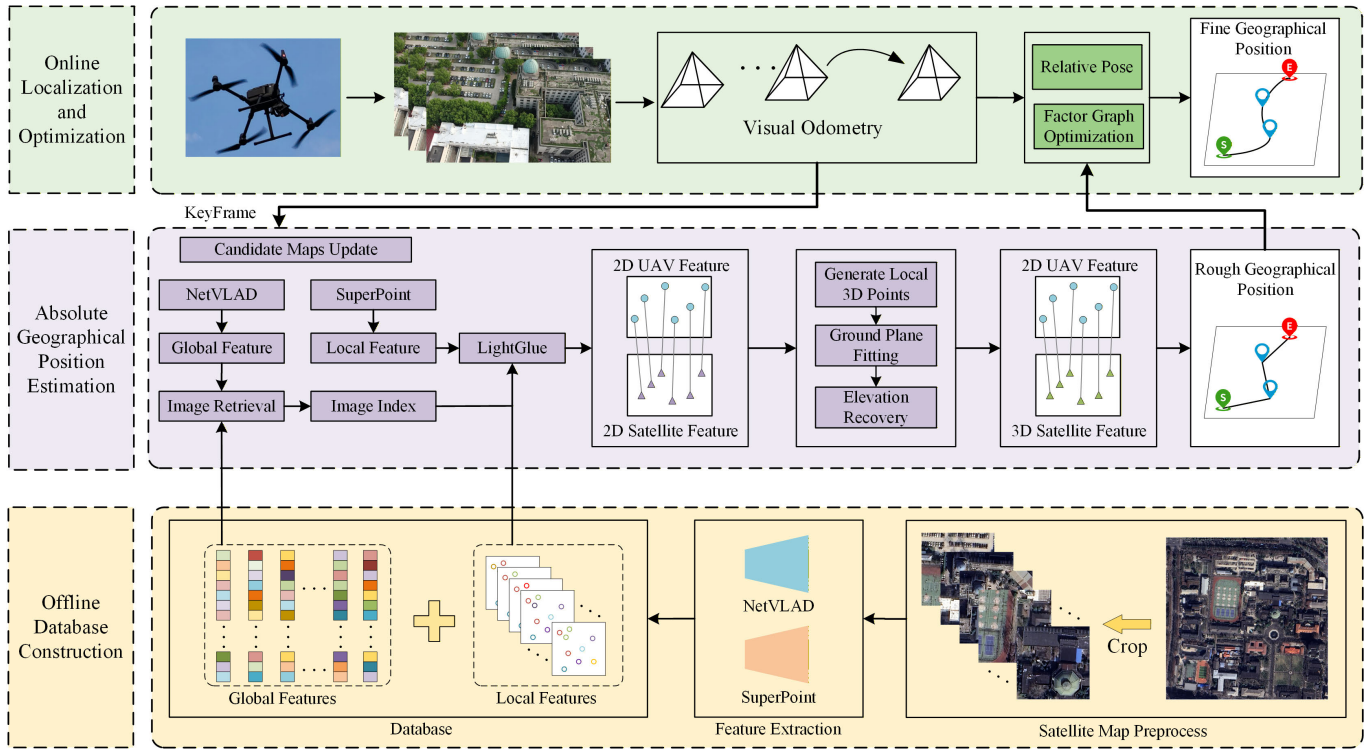


Fig. 1. Pipeline of the proposed visual geolocalization system. The system consists of an offline database construction stage and an online localization stage. In the offline stage, satellite images are processed to extract global and local features for database construction. During online localization, UAV images are used for local pose estimation via SLAM, followed by satellite image retrieval, elevation recovery, and joint pose optimization for accurate geolocalization.

methods operate in a relative coordinate frame and, without initialization based on absolute geographic coordinates, fail to provide global coordinate information. They also suffer from trajectory drift and accumulating errors during prolonged UAV flights. Although SLAM-based methods have their limitations in global localization, the constraints between keyframes and map points provide robust local positioning, which can serve as a valuable reference for geolocalization. Consequently, research efforts [13], [16] aim to overcome pure SLAM limitations by integrating the absolute geographic coordinates obtained from image matching with the relative coordinates provided by SLAM. For instance, studies such as [31] and [32] first match UAV images with satellite images to obtain absolute coordinates, and then use visual-inertial odometry (VIO) to estimate the local relative trajectory of the UAV. Finally, local BA is applied to fuse absolute and relative information for accurate geolocalization.

Whether it is direct image matching, coarse-to-fine strategies, or SLAM-integrated approaches, the aforementioned methods share a common limitation: they only use feature points from 2-D RS images. Consequently, when the UAV viewing angle is oblique, high-precision geographic coordinates cannot be obtained due to the lack of elevation information in RS images.

B. Visual Geolocalization Incorporating Satellite Image Elevation

Regardless of whether image retrieval or SLAM-based methods are used, a major limitation remains: the assumption of a nadir-viewing UAV camera, which relies solely on 2-D

feature constraints derived from satellite images [8], [14], [15]. However, maintaining a strictly downward camera pose during real-world UAV flights is often impractical due to complex environmental conditions and dynamic flights. Under oblique viewpoints, the lack of elevation information in purely 2-D satellite features leads to significant localization errors. To address this limitation, several studies [17], [18], [19] have incorporated terrain data, including DEM and DSM, to reconstruct 3-D satellite features. This allows 3-D-to-2-D matching and makes the localization process more robust under oblique UAV viewing angles. Nevertheless, DEM tends to introduce large errors in urban scenes due to their low resolution [17], which undermines their suitability for high-precision localization. Although DSM provides high-resolution elevation data, their reliance on prior data acquisition using UAVs or other platforms incurs substantial costs and significantly limits their scalability in applications.

III. METHODOLOGY

A. System Overview

As mentioned in Section I, our objective is to achieve autonomous visual geolocalization for UAVs in urban GNSS-denied environments. Therefore, the proposed positioning method relies solely on the on-board camera of the UAV and preacquired satellite imagery. The system works in two stages: offline database construction and online localization. The overall pipeline is illustrated in Fig. 1. In the offline stage, satellite images with georeferenced information are preprocessed and cropped into multiple overlapping subimages. Global features

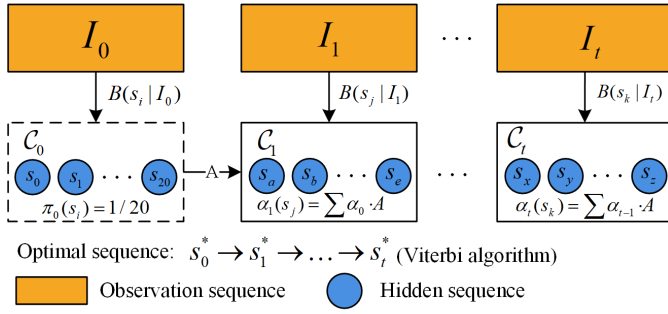


Fig. 2. Initial frame generates 20 candidate submaps C_0 through NetVLAD retrieval; subsequent frames predict the current candidate set C_t from the previous candidate set C_{t-1} using the state transition matrix $\mathbf{A}$ and filter the top 5 submaps by combining with the observation probability matrix $\mathbf{B}$; the Viterbi algorithm is used to infer the optimal submap sequence, and the optimal submap is finally determined via LightGlue local feature matching.

(NetVLAD) and local features (SuperPoint) are extracted from each subimage to build the feature database.

In the online stage, UAV image sequences are first processed by an SLAM system to estimate local relative poses. Specifically, we use ORB-SLAM3 [33] as the underlying SLAM module, leveraging its reliable monocular visual tracking to provide frame-to-frame relative pose estimates. The selected keyframes are subsequently passed to the geographical position estimation module. Within this module, the corresponding satellite subimages are efficiently retrieved from the feature database for each keyframe, using a dynamic candidate map update (DCMU) mechanism based on HMM (details in Section III-B). Next, a novel UAV coarse geographical pose estimation method considering elevation disparities in RS images is proposed, as detailed in Sections III-C and III-D1. Two-dimensional-to-three-dimensional correspondences between keyframes and satellite features are used to estimate a rough geographical position. Finally, the estimated rough geographical position of the keyframe is jointly optimized with the local relative pose using a factor graph optimization algorithm, as detailed in Section III-D2. This achieves a smooth, drift-free UAV localization that is robust against viewpoint variations.

B. DCMU Mechanism Based on HMM

To perform geolocalization of UAV images using satellite maps, this article proposes an adaptive candidate map retrieval strategy based on the HMM. Unlike conventional frame-independent retrieval strategies, our method introduces an HMM-based temporal update mechanism that dynamically refines the candidate submap set, substantially improving retrieval efficiency. Specifically, the satellite map is first divided into overlapping submap grids. In the offline phase, NetVLAD is used to extract global descriptors for each submap, which are then used to construct a global feature database, while SuperPoint is used to extract local features for fine-grained matching. During online localization, the HMM is leveraged to model UAV state transitions and observation likelihoods, enabling dynamic filtering of candidate submaps.

As illustrated in Fig. 2, this approach formulates UAV localization as a state estimation problem within an HMM

framework. The state space comprises indices of satellite submaps $\mathcal{S} = \{s_1, s_2, \dots, s_N\}$, where the state transition probability matrix $\mathbf{A}$ represents the likelihood of the UAV transitioning from submap s_i to s_j , and the observation probability matrix $\mathbf{B}$ represents the visual similarity likelihood between submap s_j and the UAV image.

For the first frame in the sequence, the system performs global retrieval using NetVLAD to obtain the top 20 submaps with the highest cosine similarity, forming the initial candidate set C_0 . The selection of “top 20” is determined by a tradeoff between retrieval accuracy and computational efficiency. The retrieval is implemented using Faiss [34], which enables efficient large-scale similarity search and supports database sizes at the million-scale level. The initial state distribution π_0 is then set as a uniform distribution over this candidate set, i.e., $\pi(s_i) = 1/20$ for $s_i \in C_0$.

For subsequent keyframes, the HMM forward algorithm predicts candidate submaps through two steps.

- 1) *State Transition Prediction*: Given the previous localization result s_{t-1} , the prior state distribution at time step t is computed using the state transition matrix $\mathbf{A}$

$$\alpha_t(s_j) = \sum_{s_i \in \mathcal{S}} \alpha_{t-1}(s_i) \cdot A(s_i, s_j) \quad (1)$$

where $\alpha_t(s_j)$ represents the probability that time t is in the submap s_j , and $A(s_i, s_j)$ encodes the spatial transition dynamics between consecutive UAV locations.

- 2) *Observation Likelihood Fusion*: For each predicted submap s_j , its NetVLAD similarity to the current UAV image I_t , denoted by $\text{sim}(s_j, I_t)$, is normalized to obtain the observation likelihood $B(s_j | I_t)$.

The Viterbi algorithm is used to infer the most probable state sequence, and the top 5 submaps with the highest posterior probabilities are selected to form the dynamic candidate set

$$C_t = \arg \max_{s_j \in \mathcal{S}} \alpha_t(s_j) \cdot B(s_j | I_t). \quad (2)$$

Subsequently, local feature matching is performed between the UAV image and each candidate submap in descending order of similarity using LightGlue. If the number of matched keypoints exceeds a predefined threshold (empirically set to 50), the corresponding submap is selected as the matching region for the current UAV frame.

The HMM-based DCMU mechanism is summarized in Algorithm 1. As a fundamental component of the proposed system, this mechanism offers a practical tradeoff between computational cost and retrieval accuracy across all the evaluated scenarios.

C. Elevation Recovery Method Based on Cross-Frame Difference Constraints

To overcome the critical limitation in UAV visual geolocalization that forces the UAV camera to maintain a strictly nadir-facing (vertically downward) orientation due to reliance on 2-D satellite imagery, we propose an elevation recovery strategy that does not depend on external elevation models such as DSM or DEM. Instead, it uses the parallax between

Algorithm 1 HMM-Based Adaptive Candidate Map Retrieval

Input: I_t : current UAV image; $D_t^{NetVLAD}$: NetVLAD descriptor of I_t ; D_t^{SP} : SuperPoint features of I_t ; $S_{library}$: satellite submaps database with precomputed NetVLAD/SP features; $\mathbf{A}$: HMM state transition matrix; $\mathbf{B}$: HMM observation probability matrix; α_{t-1} : previous state distribution (if available)

Output: S_{match} : best-matched satellite submap; α_t : updated state distribution

```

1 Function HMMCandidateMapRetrieval( $I_t, D_t^{NetVLAD}, D_t^{SP}, \mathbf{A}, \mathbf{B}, \alpha_{t-1}$ ):
2   if  $t == 0$  then
3      $C \leftarrow$  Retrieve top-20 satellite submaps from  $S_{library}$  using  $D_t^{NetVLAD}$ ;
4      $\pi \leftarrow$  Uniform distribution over submaps in  $C$ ;
5      $\alpha_0 \leftarrow \pi$ ;
6   else
7     foreach submap  $s_j \in S_{library}$  do
8        $\alpha'_t(s_j) \leftarrow \sum_{s_i \in S_{library}} \alpha_{t-1}(s_i) \cdot \mathbf{A}(s_i, s_j)$ 
9        $sim_j \leftarrow$  Cosine similarity between  $D_t^{NetVLAD}$  and  $s_j$ 's descriptor;
10       $\mathbf{B}(s_j|I_t) \leftarrow \exp(\lambda \cdot sim_j) / \sum_k \exp(\lambda \cdot sim_k)$ ;
11       $\alpha_t(s_j) \leftarrow \alpha'_t(s_j) \cdot \mathbf{B}(s_j|I_t)$ ;
12    $C \leftarrow$  Top-5 submaps with highest  $\alpha_t(s_j)$  values;
13   foreach  $S_i \in C$  do
14      $D_i^{SP} \leftarrow$  Load SuperPoint features of  $S_i$ ;
15      $matches \leftarrow$  LightGlue( $D_t^{SP}, D_i^{SP}$ ); if  $|matches| \geq threshold$  then
16       return ( $S_i, \alpha_t$ );
17   return (None,  $\alpha_t$ );

```

UAV keyframes and local triangulation to estimate the depth of feature points observed in the UAV images. By combining these depth estimates with feature correspondences established between UAV keyframes and satellite images, a 3-D geographic point set is reconstructed at real-world scale.

1) *Obtaining 2-D Correspondences Between UAV Images and Satellite Images:* Following Section III-B, the global and local features of UAV keyframes are extracted by NetVLAD and SuperPoint, respectively. The global features are used to retrieve candidate submaps from the offline database, and LightGlue is then applied to match the local SuperPoint features between the UAV image and the retrieved candidate submaps, producing the 2-D point correspondences.

2) *Correspondence Between UAV 3-D Observations and 2-D Satellite Images:* As shown in Fig. 3, for adjacent UAV keyframes F_{n-1} and F_n , we first extract SuperPoint local feature points. Then we use LightGlue to match the feature points. Based on the relative pose $\mathbf{T}$ from the SLAM system, we perform fast triangulation to obtain a sparse 3-D point cloud $\mathcal{P}^{3-D}$. For current UAV frame F_n , its local feature points

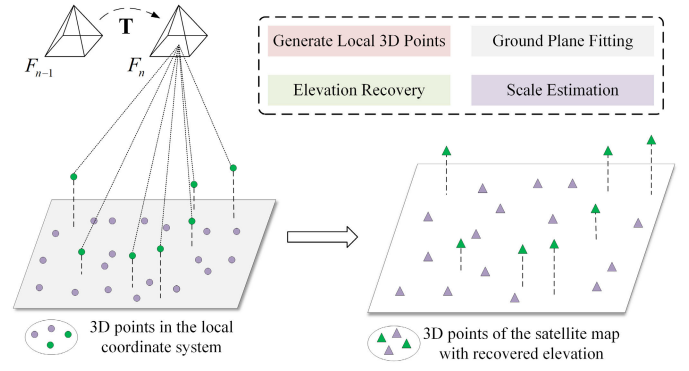


Fig. 3. Elevation recovery for satellite image feature points. The dots represent 3-D points observed by the UAV keyframes in the local coordinate system, while the triangular points denote 3-D satellite image feature points with recovered elevation information in the geographic coordinate system. Among these, the purple dots indicate points on the fit plane, and the green dots represent points with elevation information.

are

$$\mathcal{A}^{2-D} = \{A_1, A_2, \dots, A_N\}. \quad (3)$$

These feature points are linked to two different domains: the local 3-D points and the 2-D satellite points. Specifically, a subset of feature points $A_i \in \mathcal{A}^{2-D}$ are successfully triangulated using the relative pose from SLAM, resulting in corresponding 3-D points $P_i \in \mathcal{P}^{3-D}$. Meanwhile, another subset of feature points $A_j \in \mathcal{A}^{2-D}$ are matched to 2-D feature points $S_j \in \mathcal{S}^{2-D}$ on the satellite map S , as described in Section III-C1. Accordingly, we define two subsets

$$\begin{cases} \mathcal{A}_{tri}^{2-D} = \{A_i \in \mathcal{A}^{2-D} \mid A_i \langle - \rangle P_i, P_i \in \mathcal{P}^{3-D}\} \\ \mathcal{A}_{sat}^{2-D} = \{A_j \in \mathcal{A}^{2-D} \mid A_j \langle - \rangle S_j, S_j \in \mathcal{S}^{2-D}\} \end{cases} \quad (4)$$

where $\langle - \rangle$ denotes the correspondence between points. Their intersection is given by (5), which means these points are matched in both the local 3-D points and the satellite map

$$\mathcal{A}_{int}^{2-D} = \mathcal{A}_{tri}^{2-D} \cap \mathcal{A}_{sat}^{2-D}. \quad (5)$$

Therefore, for any $A_i \in \mathcal{A}_{int}^{2-D}$, the following correspondence can be obtained:

$$P_i \leftarrow A_i \rightarrow S_i \Rightarrow S_i \leftrightarrow P_i \quad (6)$$

where $P_i \in \mathcal{P}^{3-D}$, $S_i \in \mathcal{S}^{2-D}$. Finally, a set of point pairs from satellite map feature points to local 3-D points can be constructed

$$\mathcal{M}_{sat-3-D} = \{(S_k, P_k) \mid A_k \in \mathcal{A}_{int}^{2-D}\}. \quad (7)$$

This set provides essential data support for the following scale estimation and elevation recovery.

3) *Elevation Recovery:* The triangulated points $\mathcal{P}^{3-D}$ are defined in the local camera coordinate system of the UAV. Their scale and orientation cannot be directly mapped to the real-world geographic coordinate system. To align the local 3-D points with the satellite image coordinates, we first extract ground points from the 3-D point cloud.

We apply an RANSAC-based plane segmentation algorithm to the point cloud to extract the ground plane. Specifically,

given a point cloud $\mathcal{P}$, the algorithm fits a plane model defined by the equation

$$ax + by + cz + d = 0 \quad (8)$$

where a , b , c , and d are the plane parameters. The RANSAC algorithm iteratively selects minimal subsets of points to hypothesize candidate planes and identifies the plane that has the most inlier points within a predefined distance threshold. The set of inliers corresponds to points belonging to the ground plane.

Then we can compute the distance from each point $P_i = (x_i, y_i, z_i)$ in the point cloud to the estimated plane using the following equation:

$$h_i = -\frac{ax_i + by_i + cz_i + d}{\sqrt{a^2 + b^2 + c^2}}. \quad (9)$$

Points with absolute distance less than a predefined threshold ε , namely, those satisfying $|h_i| < \varepsilon$, are identified as inliers and collected into the subset $\mathcal{G}$, which we consider to represent the ground plane. In our implementation, ε is set to 0.3, which is consistent with the noise level typically observed in triangulation-based 3-D reconstruction from sequential UAV images and was empirically validated to provide a good balance between retaining true ground points and suppressing outliers.

For each point $P_i \in \mathcal{G}$, we can associate a corresponding geographic coordinates $(x_i^{\text{geo}}, y_i^{\text{geo}})$ on the satellite map through (7). The pixel coordinates of a point on the satellite map (x_i, y_i) can be converted into geographic coordinates $(x_i^{\text{geo}}, y_i^{\text{geo}})$ using the following equation:

$$\begin{cases} x^{\text{geo}} = x_0^{\text{geo}} + r_x \cdot x_i \\ y^{\text{geo}} = y_0^{\text{geo}} + r_y \cdot y_i \end{cases} \quad (10)$$

where $(x_0^{\text{geo}}, y_0^{\text{geo}})$ denotes the geographic coordinates of the top-left corner of the satellite map, and r_x and r_y represent the horizontal and vertical ground resolutions of the map, respectively.

To estimate the scale factor between the local 3-D coordinate system and the geographic coordinate system, we first randomly sample pairs of points (i, j) from the subset $\mathcal{G}$. For each pair, we compute the Euclidean distances both in the local 3-D coordinate system using (11) and in the geographic coordinate system using (12)

$$d_l(i, j) = \|P_i^{3-D} - P_j^{3-D}\| \quad (11)$$

$$d_g(i, j) = \|(x_i^{\text{geo}}, y_i^{\text{geo}}) - (x_j^{\text{geo}}, y_j^{\text{geo}})\|. \quad (12)$$

Using these paired distances, we construct a set of scale factors $\mathcal{S}$

$$\mathcal{S} = \left\{ \frac{d_g(i, j)}{d_l(i, j)} \right\} \quad (13)$$

which represent candidate scale factors between the two coordinate systems. To reduce the influence of outliers, we apply a trimmed mean operation $\text{trim}_k(\cdot)$ to $\mathcal{S}$, removing the extreme values before averaging. In our implementation, the parameter k is set to 1% of the size of the scale set $\mathcal{S}$. This choice is based on our empirical observation that the most significant outliers consistently occur in the lowest and highest 1% of

the scale ratios. By discarding only these extreme portions, we preserve the majority of valid scale estimates while effectively suppressing outlier influence. The final *scale* is then calculated by the following equation:

$$\text{scale} = \frac{1}{|\mathcal{S}'|} \sum_{s \in \mathcal{S}'} s, \quad \mathcal{S}' = \text{trim}_k(\mathcal{S}). \quad (14)$$

For each pair $(S_k, P_k) \in \mathcal{M}_{\text{sat-3-D}}$, we compute the elevation of S_k as

$$z_k = \text{scale} \cdot h_k \quad (15)$$

where h_k is obtained via (9), and it represents the distance from the local 3-D point P_k , corresponding to S_k , to the ground plane.

As a result, we obtain the final set of 3-D feature points on the satellite map

$$\mathcal{S}^{3-D} = \{(x_i^{\text{geo}}, y_i^{\text{geo}}, z_i)\}. \quad (16)$$

This process enables the transformation of 2-D satellite feature points into 3-D feature points, thereby providing robust geometric constraints to support the subsequent localization tasks.

D. Geographic Pose Estimation and Optimization

To address the limitation that most existing SLAM-Geo fusion methods rely only on 2-D geographic coordinates, we introduce a unified 3-D geospatial fusion framework that couples the recovered 3-D geographic coordinates with the local SLAM coordinate system. Unlike conventional approaches that treat SLAM poses and geographic information as loosely connected constraints, our method tightly integrates three complementary sources of information: the local relative poses estimated by SLAM, the absolute geographic poses derived from our elevation-augmented 3-D satellite features, and a dynamically optimized Sim(3) transformation that continuously eliminates redundant global scale drift. This formulation enables the system to achieve globally consistent and drift-free UAV geolocation.

1) *Geographic Pose Estimation:* Through the above steps, the 2-D satellite feature point set $\mathcal{S}^{2-D}$ is extended into a 3-D point set $\mathcal{S}^{3-D}$ by assigning elevation values. Each 3-D point in $\mathcal{S}^{3-D}$ has an associated geographic coordinate, which can be paired with a corresponding 2-D point in the UAV image. This enables the formulation of a perspective-n-point (PnP) problem for camera pose estimation. An initial coarse pose $\mathbf{T}$ is estimated using the direct linear transform (DLT) algorithm. This provides a preliminary rotation $\mathbf{R}$ and translation $\mathbf{t}$ of the UAV camera relative to the geographic coordinate system.

To improve the accuracy, a nonlinear-least squares optimization is then performed by minimizing the total reprojection error of the 3-D satellite points. The objective function e is defined as

$$e = \min_{\mathbf{R}, \mathbf{t}} \sum_{i=1}^n \|\mathbf{z}_i - \mathbf{\Pi}(\mathbf{R}\mathbf{P}_i + \mathbf{t})\|^2 \quad (17)$$

where $\mathbf{P}_i \in \mathcal{S}_{3-D}$ are the 3-D satellite points, $\mathbf{z}_i$ are the corresponding 2-D image observations, and $\mathbf{\Pi}(\cdot)$ denotes the perspective projection function under known camera intrinsics.

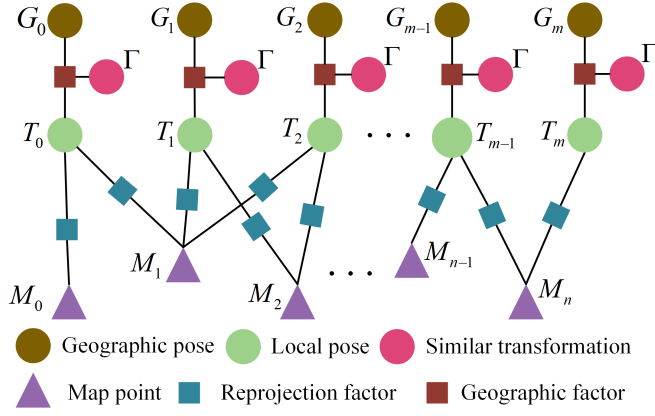


Fig. 4. Factor graph representation of our system.

The optimization is solved with the Levenberg–Marquardt algorithm, resulting in a better UAV pose in the geographic coordinate system.

2) *Pose Optimization*: Solving PnP independently for each keyframe ignores the geometric constraints between keyframes, which can degrade the accuracy of the localization and cause unstable trajectory jumps. To address this issue, we propose a joint optimization method that fuses absolute geographic poses with local SLAM poses. By integrating the relative pose estimations from SLAM and the absolute pose estimations derived from 3-D satellite map features, we can achieve a smoother and more continuous trajectory in the geographic coordinate system and improve localization accuracy. In addition, the introduction of absolute pose constraints in the geographic coordinate system helps mitigate the long-term drift commonly observed in SLAM systems.

Since monocular ORB-SLAM3 lacks real-world scale information, we introduce a scale correction mechanism during the initialization phase of the SLAM module. Specifically, for the keyframes used in initialization, denoted as KF_{ref} and KF_{cur} , we estimate their positions in the geographic coordinate system, denoted as p_1^{geo} and p_2^{geo} , and compute their Euclidean distance as the reference scale. We also extract their corresponding positions in the local SLAM coordinate as p_1 and p_2 . The scale correction factor is defined as

$$s = \frac{\|p_2 - p_1\|}{\|p_2^{\text{geo}} - p_1^{\text{geo}}\|}. \quad (18)$$

The application of this scale factor during SLAM initialization brings the local scale closer to that of the real world, which helps improve stability during subsequent processing.

We further design a local backend optimization algorithm based on pose fusion. In this process, the estimated geographic poses for keyframes are introduced as constraints in the local bundle adjustment. Let $T_i \in \text{SE}(3)$ denote the pose of the i th keyframe in the SLAM coordinate system, while $G_i \in \text{Sim}(3)$ represents its corresponding observation in geographic coordinates. The transformation from the SLAM coordinate system to the geographic coordinate system is represented by a similarity transformation $\Gamma \in \text{Sim}(3)$. As shown in Fig. 4, we define three types of variable nodes in the factor graph: keyframe poses $T_i \in \text{SE}(3)$, map point positions $M_j \in \mathbb{R}^3$,

and the transformation $\Gamma \in \text{Sim}(3)$. Two types of residuals are defined in the optimization: visual reprojection residual and geographic pose residual.

Based on the pinhole camera model, the visual residual e_{ij}^{vis} is defined as

$$e_{ij}^{\text{vis}} = z_{ij} - \Pi(T_i, M_j) \quad (19)$$

where e_{ij}^{vis} denotes the reprojection residual of the 3-D map point M_j with respect to the UAV camera pose T_i at the i th keyframe. $z_{ij} = [u_s, v_s]^T \in \mathbb{R}^2$ represents the observed pixel coordinates of the feature point. The function $\Pi(\cdot)$ reprojects the 3-D point M_j onto the image plane using the camera projection model parameterized by the extrinsic pose T_i .

For keyframes with geographic pose observations, to express the SLAM pose T_i in the same transformation space as the geographic observations, we transform the SLAM pose T_i into a unit-scale similarity transformation, denoted as $S_i = \text{Sim}(T_i, 1)$. To remove the effect of redundant global scale in the error formulation, we further introduce a scale-only $\text{Sim}(3)$ transformation $S^{-1} = \text{Sim}(\mathbf{I}, 0, s_0)^{-1}$, where s_0 denotes the scale component extracted from the current estimate of the similarity transformation $\Gamma \in \text{Sim}(3)$. Given the geographic observation $G_i \in \text{Sim}(3)$, the residual is defined as the logarithmic map of the relative transformation between the transformed SLAM pose and the observation

$$e_i^{\text{geo}} = \log(G_i^{-1} \Gamma S_i S^{-1}). \quad (20)$$

By minimizing this residual, the similarity transformation Γ from the SLAM coordinate system to the geographic coordinate system is dynamically optimized. Importantly, Γ contains a scale component that is continuously refined during the optimization. As new geographic constraints are incorporated along the trajectory, the scale term within Γ is repeatedly updated, allowing the system to actively compensate for scale drift rather than relying solely on the initialization stage. In this manner, both the global scale and the positional offsets are jointly corrected throughout the factor-graph optimization. This allows all the SLAM-estimated poses to be consistently aligned with the global geographic coordinate system, enabling accurate geolocation across the entire trajectory.

Therefore, for m keyframes with geographic pose observations and their corresponding n map points, the overall cost function is defined as

$$\mathcal{E} = \sum_{i=1}^m \left\{ \sum_{j=1}^n \rho(e_{ij}^{\text{vis}} \mathbf{W}_i^{\text{vis}} e_{ij}^{\text{vis}}) + \alpha_i e_i^{\text{geo}T} \mathbf{W}_i^{\text{geo}} e_i^{\text{geo}} \right\} \quad (21)$$

where $\rho(\cdot)$ denotes the Huber robust kernel function, which can suppress the influence of potentially large errors in geographic pose observations. $\mathbf{W}_i^{\text{vis}}$ and $\mathbf{W}_i^{\text{geo}}$ represent the information matrices for the visual reprojection residual and the geographic residual, respectively. To further enhance the robustness of the geographic pose constraints, we introduce an adaptive weighting coefficient α_i into the geographic residual

$$\alpha_i = \frac{N_i}{\bar{N}}, \quad \bar{N} = \frac{1}{m} \sum_{k=1}^m N_k \quad (22)$$

where N_i denotes the number of UAV–satellite matches associated with the i th keyframe and $\bar{N}$ is their average over

all frames. This adaptive scheme automatically downweights unreliable geographic observations. Consequently, the factor graph becomes more resilient to fluctuations and noise in geographic pose estimation. This backend optimization method yields not only globally referenced poses but also smooth and stable UAV trajectories.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we present the experimental results of the proposed method on the RealUAV [35] and SEULoc datasets. Among them, SEULoc is a dataset collected by us in real-world scenarios. Fig. 5 shows three experimental scenarios selected from the RealUAV dataset. We compare our method against both the absolute and relative localization methods. The absolute localization method, referred to as ElevPnP, uses our proposed elevation recovery algorithm to estimate the elevation of feature points extracted from satellite images, thereby constructing a 3-D point set. The UAV pose is then estimated by solving the PnP problem using these 3-D satellite feature points. For the relative localization method, we adopt the classic SLAM algorithm ORB-SLAM3 (monocular mode). To align the SLAM-estimated trajectory with the correct scale and the geographic coordinate system, we use ElevPnP to estimate the geographic coordinates of the first two keyframes during the initialization stage. A scale correction is then applied based on the geodetic distance between these keyframes. All the localization results are transformed into the Universal Transverse Mercator (UTM) coordinate system for evaluation and comparison. Furthermore, to better demonstrate the effectiveness of the proposed elevation recovery algorithm in improving localization accuracy, we perform additional comparisons with a traditional homography-based method on oblique view data from the SEULoc dataset. All the experiments are conducted on a computer equipped with an Intel Core i9-13900HX CPU, 32GB DDR5 RAM, and an NVIDIA RTX 4060 Laptop GPU.

A. Metrics

To comprehensively evaluate the performance of different visual localization methods, we adopt the open-source benchmarking tool *evo* [36], which calculates the absolute trajectory error (ATE) by comparing the estimated UAV trajectory with the ground truth. To provide a detailed statistical analysis of localization accuracy, five key metrics are derived from the ATE distribution: maximum error (Max), minimum error (Min), mean error, root mean square error (RMSE), and standard deviation (Std). These metrics offer a robust and comprehensive evaluation of the performance of various visual localization methods.

B. Evaluation of DCMU

Retrieving the corresponding satellite image for a given UAV query image from a large satellite image database is a prerequisite for visual geolocalization. Fig. 6 presents a comparison between our proposed DCMU mechanism based on HMM and direct retrieval. In the visualization, the query UAV images are displayed on the left side of the dashed line, while the most similar satellite images, sorted in descending

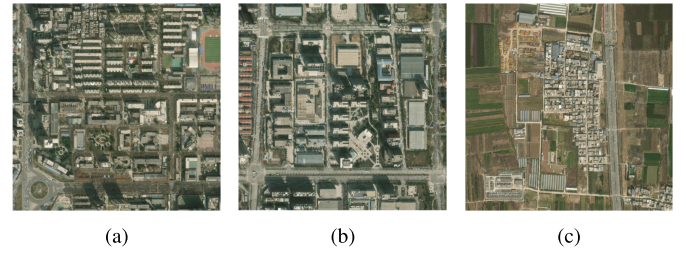


Fig. 5. Satellite images of the regions used in our localization experiments. (a) and (b) City1 and City3 scenes in the RealUAV dataset, respectively, and (c) Village1 scene.



Fig. 6. Comparison of satellite image retrieval results using our DCMU strategy versus direct retrieval. Purple boxes indicate the query UAV images, green boxes denote correctly matched satellite images, and red boxes highlight mismatched satellite images.

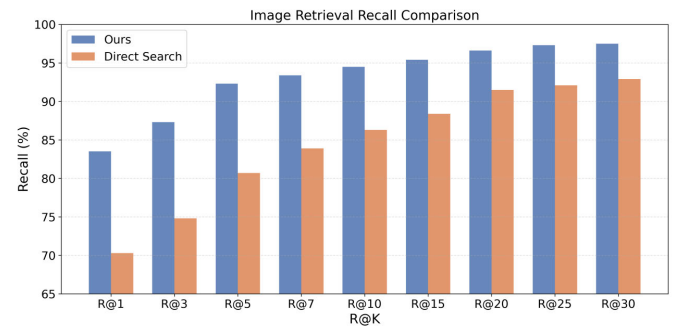


Fig. 7. Comparison of recall rates between our retrieval method and direct retrieval.

order of similarity based on the NetVLAD descriptor, are shown on the right side. As shown in the figure, direct retrieval tends to produce more incorrect matches, requiring additional computational resources to filter out false positives before arriving at the correct satellite image. In contrast, our method explicitly models the UAV's motion dynamics through the HMM, leveraging prior state transitions and observation probabilities to dynamically prune the candidate space. This results in significantly more accurate retrieval with fewer mismatches, as the algorithm adaptively focuses on regions with higher likelihood based on sequential image analysis.

In addition, a quantitative comparison was conducted between the HMM-based retrieval performance of our proposed DCMU module and the direct retrieval method

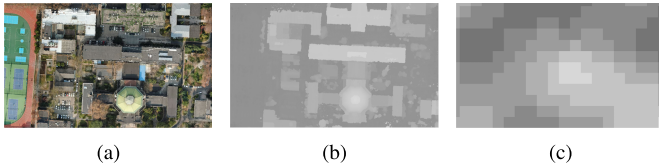


Fig. 8. Local comparison of DSM and DEM over the UAV flight area in the SEULoc dataset. (a) Local RGB image. (b) DSM generated using DJI Terra. (c) ALOS DEM with a spatial resolution of 12.5 m. Compared with the DEM, the DSM exhibits substantially higher geometric fidelity and is adopted as the ground-truth elevation reference for the satellite imagery.

TABLE I

COMPARISON OF ELEVATION ESTIMATION ACCURACY BETWEEN THE PROPOSED METHOD AND DEM DATA

Metric	SEULoc01		SEULoc02		SEULoc03	
	DEM	Proposed	DEM	Proposed	DEM	Proposed
Max(m)	48.35	11.32	31.68	12.39	48.09	14.02
Min(m)	0.01	0.04	0.01	0.13	0.02	0.09
Mean(m)	11.28	1.99	12.75	2.23	13.70	2.13
RMSE(m)	13.82	2.50	14.38	2.79	16.08	2.67

commonly adopted in most existing studies. Fig. 7 presents the comparative results of the two methods in terms of recall rate. The evaluation uses the R@K metric, which denotes the probability that the correct matching satellite submap is included among the top-K retrieved results. As illustrated in the figure, our retrieval method achieves a substantial improvement in retrieval performance compared with direct retrieval. Particularly when the number of returned candidates is small, the performance gain is more pronounced—this advantage further contributes to reducing computational resource consumption.

To ensure a focused evaluation of the proposed joint back-end optimization algorithm and the integration of restored elevation information from satellite images, DCMU is used as a foundational step in the following experiments.

C. Evaluation of Elevation Recovery Accuracy

To evaluate the effectiveness of our elevation recovery module, an elevation error analysis was conducted. Since high-precision elevation ground truth is difficult to obtain and commonly available DEM products provide a spatial resolution of only 12.5 m at best, a high-resolution DSM over the SEULoc dataset area was generated using DJI Terra and adopted as the ground-truth elevation reference. Its geometric accuracy reaches the centimeter level. As illustrated in Fig. 8, the DSM we produced exhibits significantly finer structural detail than the 12.5-m ALOS DEM.

For all the three sequences in SEULoc, the elevations of the matched satellite-image feature points recovered by our method were recorded and compared against the ground truth. In addition, to more comprehensively verify the effectiveness of the proposed elevation recovery approach, the DEM-based elevation of each feature point was also compared with the ground truth. The elevation error statistics for all the three sequences are summarized in Table I. It can be observed that compared with the low-resolution DEM, the elevations estimated by our method are much closer to the ground truth. This improvement is attributed to the effective use of fine-

grained feature correspondences across UAV frames, which provide depth cues, and the subsequent matching between UAV frames and satellite images, enabling more accurate elevation estimation for satellite-image feature points.

Furthermore, Fig. 9 presents a visualization of the elevation estimates for a representative frame in the SEULoc01 sequence. Compared with the DEM-based elevations, the elevations predicted by our method more accurately follow the distribution of the actual terrain and exhibit smaller overall errors.

D. Experiments on RealUAV

1) *Dataset Introduction:* RealUAV is an open-source dataset designed for UAV-based visual geolocalization tasks. The UAV images are captured at a fixed altitude of 200 m with a resolution of 490×490 pixels. During data collection, the UAV flies at a speed of 6 m/s, and videos are recorded at 60 fps. Images are sampled from the video at intervals of every 15 frames. The corresponding satellite RS images are obtained from Google Maps at zoom level 19, with a ground resolution of 0.247 m per pixel. In this article, the original satellite maps are divided into patches of size 768×768 pixels using a sliding window approach with a stride of 150 pixels. Each patch is annotated with precise geographic coordinates to support accurate localization. We select three representative scenes for our experiments: two urban scenes (City1 and City3) and one rural scene (Village1). These scenes contain a total of 1101 UAV images and cover approximately 5351.17 m of flight distance.

2) *Results and Analysis:* To better compare the performance of different methods, we set the geographic position of the first frame of ORB-SLAM3, estimated by ElevPnP, as the starting point of its trajectory. In this way, the trajectories estimated by all the methods are aligned in the same geographic coordinate system for comparison. The UAV trajectories estimated by the different methods are presented in Figs. 10–12. It can be seen that the method proposed in this article performs in general better than the other methods. Without external geographic constraints, the trajectory estimated by ORB-SLAM3 drifts over time, resulting in increasing deviation from the ground truth. On the other hand, ElevPnP shows sudden jumps in the trajectory, indicating instability in continuous localization. In contrast, our method produces a smoother trajectory and effectively suppresses long-term drift by jointly optimizing absolute geographic poses and local relative poses from SLAM.

Table II presents the localization results of four methods on the RealUAV dataset. It can be observed that the proposed method achieves the best performance across all the evaluation metrics. Due to the accumulation of drift during long-term operation, ORB-SLAM3 yields significantly higher errors in all the metrics except for the minimum error (Min), compared with both ElevPnP and the proposed method. In comparison to the SAVL method, our approach achieves lower positions errors across the three representative scenarios. This improvement is attributed to the fact that SAVL relies solely on 2-D features extracted from satellite imagery, whereas our method estimates the elevation information of the satellite

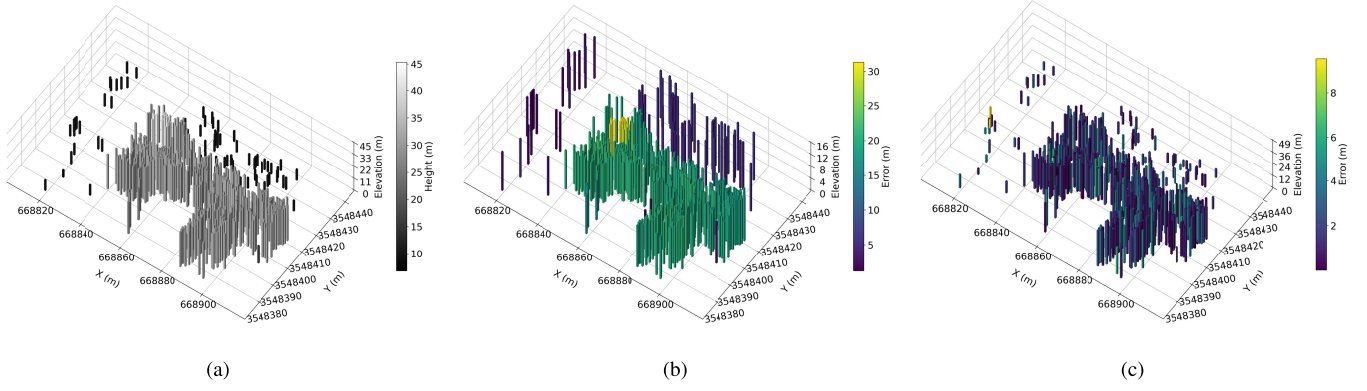


Fig. 9. Visualization of elevation information for the matched satellite-image feature points of a representative frame from the SEULoc01 sequence. (a) Ground-truth DSM elevation. (b) Elevation from the ALOS DEM. (c) Elevation estimated by the proposed method. It can be observed that the elevations estimated by our method better follow the distribution of the actual terrain and exhibit smaller overall errors.

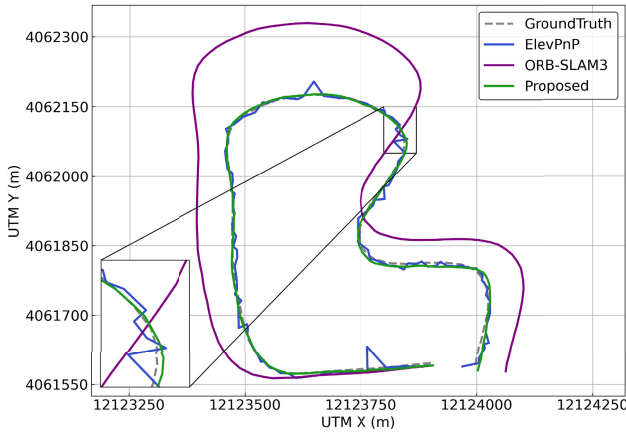


Fig. 10. UAV trajectories estimated by different methods in City1 scene.

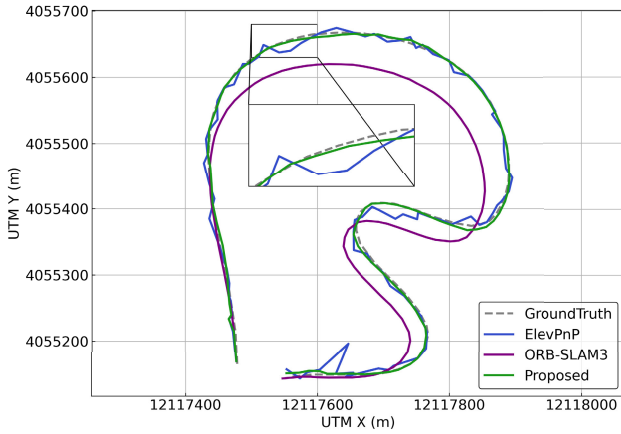


Fig. 11. UAV trajectories estimated by different methods in City3 scene.

images and incorporates a joint optimization with the relative poses, thereby delivering superior performance.

As a single-frame localization method, ElevPnP tends to exhibit sudden jumps in estimated positions. This results in large maximum errors, reaching 50.65 and 56.48 m in the City1 and City3 scenarios, respectively. In the newly Village1 scenario, ElevPnP also produces a considerable maximum error of 33.91 m. In contrast, the proposed method incorporates local pose constraints from the UAV, effectively

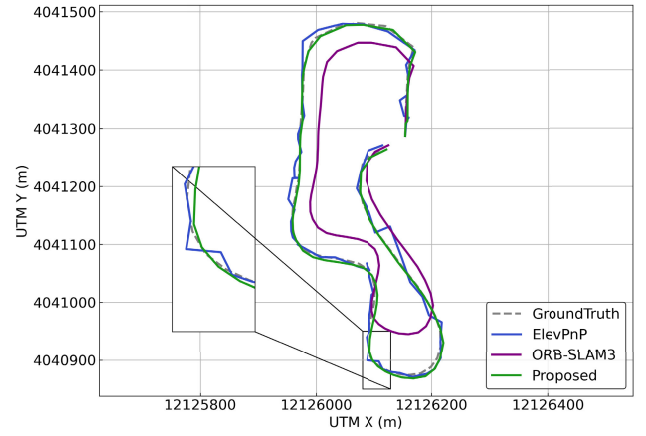


Fig. 12. UAV trajectories estimated by different methods in Village1 scene.

TABLE II
LOCALIZATION PERFORMANCE COMPARISON ON THE REALUAV DATASET

Scene	Method	Max(m)	Min(m)	Mean(m)	RMSE(m)	Std(m)
City1	ORB-SLAM3	158.75	5.98	92.46	101.55	41.99
	ElevPnP	50.65	0.86	7.65	10.02	6.48
	SAVL	-	-	8.71	-	-
	Proposed	21.03	0.76	6.21	7.48	4.12
City3	ORB-SLAM3	53.33	2.25	34.79	37.66	14.42
	ElevPnP	56.48	0.55	8.48	11.01	7.03
	SAVL	-	-	8.71	-	-
	Proposed	19.68	0.29	6.36	7.80	4.52
Village1	ORB-SLAM3	70.73	1.16	39.33	43.21	17.90
	ElevPnP	33.91	1.33	6.99	9.50	6.42
	SAVL	-	-	8.71	-	-
	Proposed	11.38	1.30	5.92	6.17	2.75

suppressing such outlier jumps. As a result, the maximum localization errors are reduced to 21.03 and 19.68 m in City1 and City3, respectively, and further to 11.38 m in Village1. Moreover, our method achieves notable improvements in mean error and RMSE. Compared with ElevPnP, the proposed method reduces the mean error and RMSE by 18.8% and 25.3% in City1, and by 25% and 29.16% in City3, respectively, while in Village1 the mean error and RMSE are reduced by 15.3% and 35.1%, respectively. These results demonstrate that the joint optimization of absolute geographic poses and local relative poses not only suppresses unstable estimates

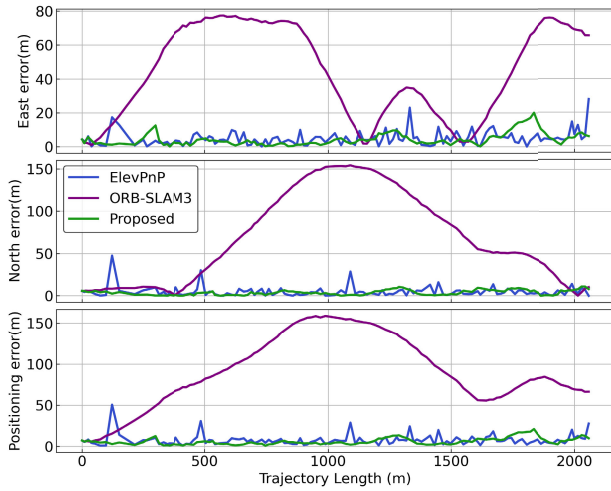


Fig. 13. Positioning errors of different methods in City1 scene.

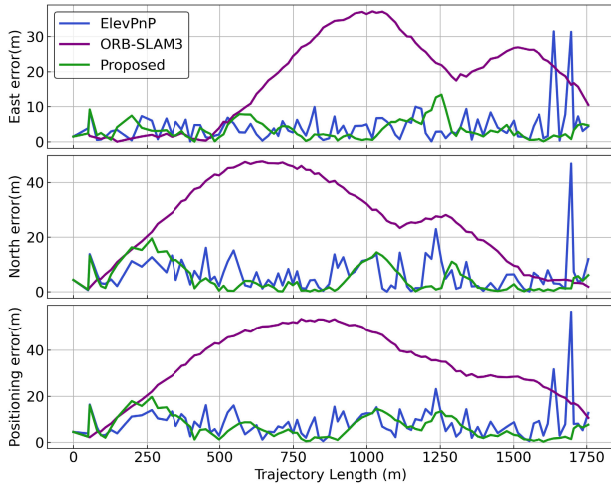


Fig. 14. Positioning errors of different methods in City3 scene.

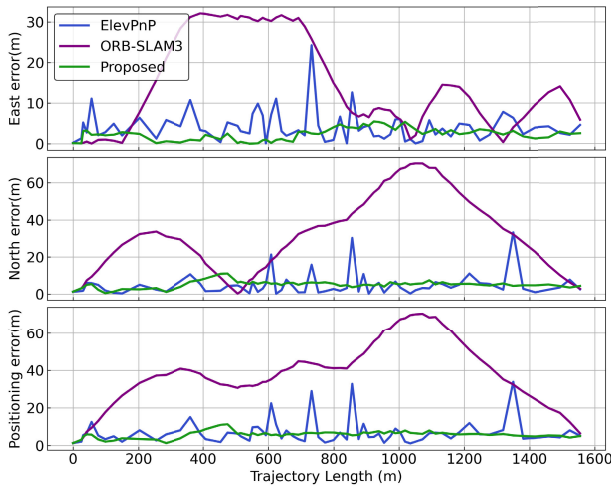


Fig. 15. Positioning errors of different methods in Village1 scene.

but also significantly improves overall localization accuracy. Figs. 13–15 illustrate the evolution of East, North, and overall positioning errors along the entire flight trajectories in the

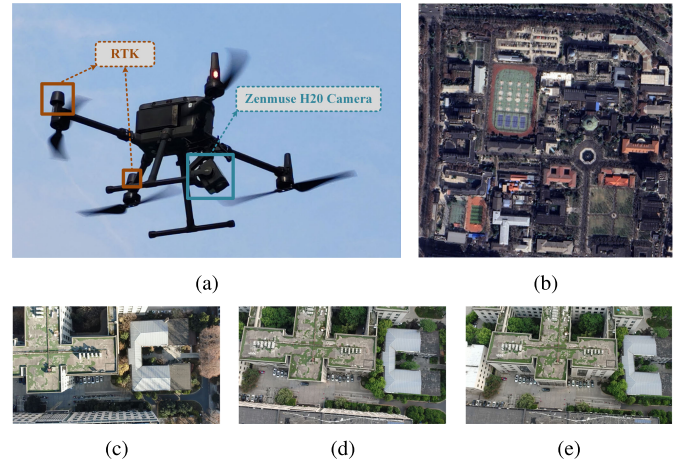


Fig. 16. Data collection platform and scenes of the SEULoc dataset. (a) UAV used for data acquisition. (b) Satellite map of the UAV flight area. (c)–(e) Images captured with camera pitch angles of -90° (vertically downward), -80° , and -70° , respectively.

TABLE III
DETAILED SPECIFICATIONS OF THE SEULOC DATASET

Sequence	Pitch($^\circ$)	Resolution(pixel)	Length(m)	Altitude(m)
SEULoc01	-90	1920*1080	846.45	105
SEULoc02	-80	1920*1080	591.8	110
SEULoc03	-70	1920*1080	583.31	110

City1, City3, and Village1 scenes. It can be seen that the proposed method significantly reduces the localization error compared with ORB-SLAM3 and does not exhibit the abrupt error spikes observed in ElevPnP.

E. Experiments on SEULoc

1) *Dataset Introduction:* To further assess the localization performance of the proposed method and the improvements introduced by recovering elevation information from satellite images, we collected a real-world dataset named SEULoc. Unlike the RealUAV dataset, SEULoc consists of images captured under oblique UAV viewing angles in low-altitude urban environments, which better reflect realistic flight conditions. Specifically, the dataset includes three subsets with data collected across different seasons and times: SEULoc01 was acquired in winter at 09:14, while SEULoc02 and SEULoc03 were captured in summer at 07:10 under overcast conditions. The detailed specifications of the SEULoc dataset are provided in Table III, and the data acquisition setup and experimental scenes are illustrated in Fig. 16. During the data collection process, we used a DJI M300 RTK UAV equipped with a Zenmuse H20 camera. The UAV recorded video sequences at 30 Hz with a resolution of 1080p. To construct the localization dataset, we uniformly sampled one frame every three frames from the recorded videos. The UAV's positions recorded by the RTK system with centimeter-level accuracy were used as ground truth. Satellite images were obtained from Google Earth with a ground resolution of 0.3 m per pixel. The satellite map was divided into 400×400 pixel patches using a sliding window approach with a stride of 100 pixels to construct the reference database.

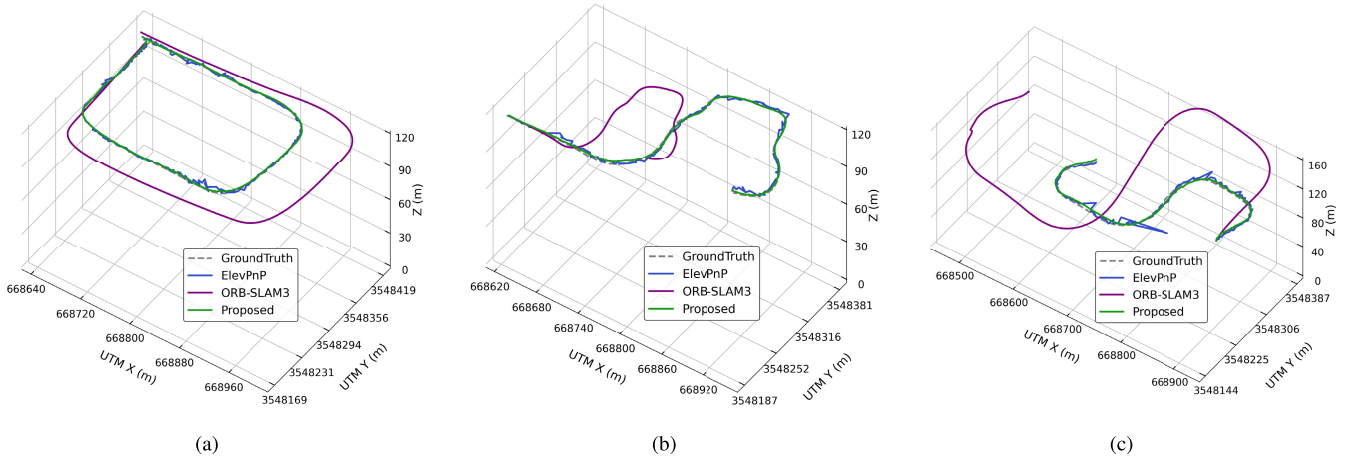


Fig. 17. Trajectories estimated by different methods in the SEULoc dataset. (a) Sequence SEULoc01. (b) Sequence SEULoc02. (c) Sequence SEULoc03.

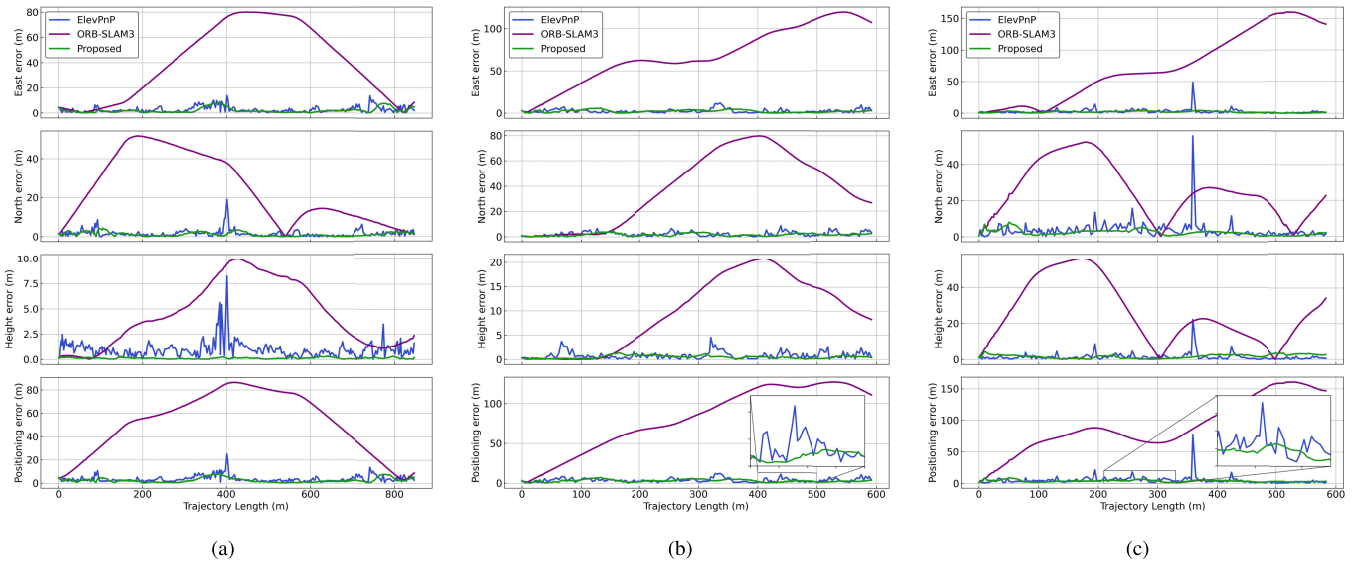


Fig. 18. Positioning errors of different methods in the SEULoc dataset. (a) Sequence SEULoc01. (b) Sequence SEULoc02. (c) Sequence SEULoc03.

2) *Results and Analysis*: Fig. 17 illustrates the estimated 3-D trajectories for three sequences in the SEULoc dataset using different methods. Meanwhile, Fig. 18 presents the localization error curves of each method across the three sequences, including the overall localization error, the northward error, the eastward error, and the altitude error. It is observed that as the UAV camera pitch angle varies, both the ElevPnP method, which uses satellite map feature points with recovered elevation, and the proposed method maintain estimated trajectories close to the ground truth without notable deviations. In contrast to methods that rely solely on vertically downward UAV camera data, recovering the elevation of satellite map feature points expands the range of applicable scenarios for UAV localization under GNSS-denied conditions. This better reflects real-world conditions, where the UAV camera cannot consistently remain vertically downward. Table IV presents the localization results of different methods on the SEULoc dataset, showing that the proposed method achieves high localization accuracy despite changes in UAV camera pitch angle, performing best across most metrics.

TABLE IV
COMPARISON OF LOCALIZATION RESULTS ON THE SEULoc DATASET

Sequence	Method	Max(m)	Min(m)	Mean(m)	RMSE(m)	Std(m)
SEULoc01	ORB-SLAM3	86.55	3.13	49.31	56.24	27.04
	ElevPnP	24.91	0.51	3.60	4.56	2.80
	Proposed	8.13	0.13	2.48	3.14	1.93
SEULoc02	ORB-SLAM3	126.99	0.44	80.93	90.72	41.01
	ElevPnP	12.13	0.19	3.68	4.44	2.48
	Proposed	7.11	0.20	2.84	3.21	1.50
SEULoc03	ORB-SLAM3	160.47	1.43	88.64	100.24	46.82
	ElevPnP	77.32	0.62	4.58	7.75	6.25
	Proposed	8.97	1.43	2.88	3.68	1.39

Similar to the results obtained from the RealUAV dataset, ORB-SLAM3, which lacks external absolute pose constraints, exhibits large localization errors over long-distance trajectories. In contrast, both ElevPnP and the proposed method eliminate cumulative drift, resulting in significantly improved performance compared with ORB-SLAM3. However, ElevPnP still suffers from sudden jumps in localization results—most

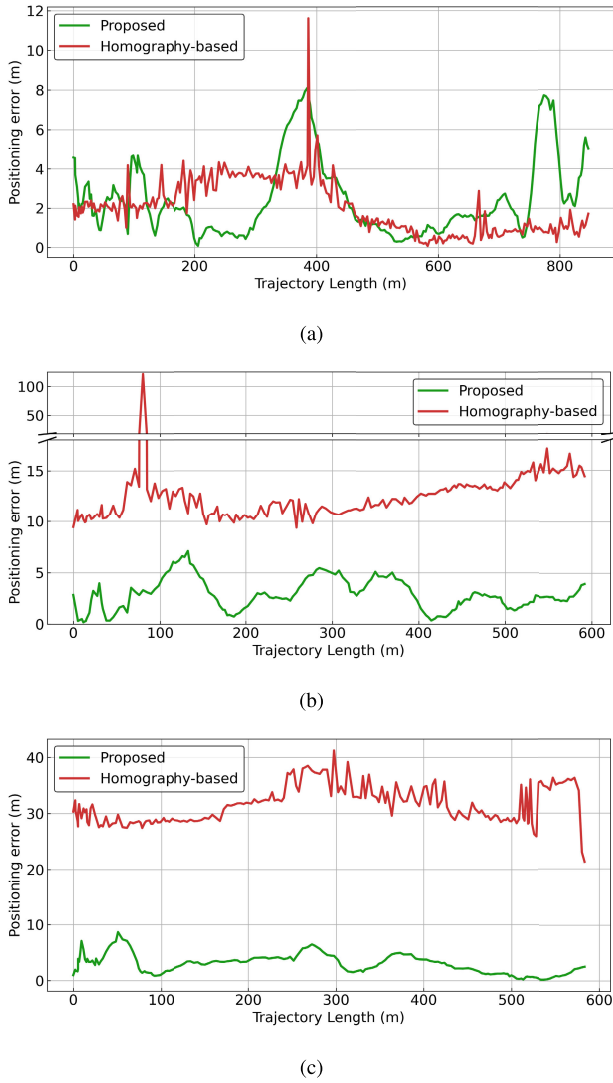


Fig. 19. Positioning errors of the proposed method and the homography-based method on the SEULoc dataset under varying camera pitch angles. (a)–(c) UAV pitch angles of -90° , -80° , and -70° , respectively.

notably in the SEULoc03 sequence, where the maximum localization error reaches 77.32 m. Moreover, the proposed method achieves much lower Std than ElevPnP, particularly in the SEULoc03 sequence, where the Std drops from 6.25 to 1.39 m. This indicates that the proposed method not only enhances localization accuracy but also provides greater stability during continuous UAV localization.

In addition, to further validate the improved localization accuracy and robustness of recovering the elevation of the feature point of the satellite map, we compared it with a method based on the homography matrix. This method treats both the UAV-captured images and satellite maps as planar surfaces, computing the homography matrix H from matched feature points between the UAV image and satellite map. The UAV image center's position on the satellite map, derived from H , serves as the UAV's geographic coordinates.

Table V displays the localization results of both the methods under different pitch angles, while Fig. 19 shows the localization error curves. It is noted that when the UAV camera is is

TABLE V
COMPARISON OF LOCALIZATION ACCURACY (MEAN AND RMSE) BETWEEN HOMOGRAPHY-BASED AND PROPOSED METHODS

Sequence	Pitch ($^\circ$)	Mean (m)		RMSE (m)	
		H-based	Proposed	H-based	Proposed
SEULoc01	-90	2.01	2.48	2.44	3.14
SEULoc02	-80	12.96	2.84	12.54	3.21
SEULoc03	-70	31.51	3.68	31.67	3.93

H-based refers to the Homography-based method.

TABLE VI
AVERAGE PER-FRAME RUNTIME OF EACH MODULE ON THE SEULoc AND REALUAV DATASETS

Module	Runtime (SEULoc)	Runtime (RealUAV)
DCMU	0.063s	0.041s
SP + LG	0.106s	0.057s
Elevation Recovery	0.024s	0.019s
Pose Estimation and Optimization	0.070s	0.018s
Total	0.263s	0.135s

vertically downward (pitch angle of -90°), the homography-based method slightly outperforms the proposed method. This is because the estimated elevation in our method may not fully match the true elevation, introducing some error in absolute geopose estimation. However, as the UAV camera pitch angle changes from -90° to -70° , the mean error and RMSE of the homography-based method increase significantly, with the mean error rising from 2.01 to 31.51 m and the RMSE increasing from 2.44 to 31.67 m. In contrast, the proposed method maintains relatively stable performance, exhibiting only a slight increase in both the metrics. These results indicate that the homography-based method becomes unreliable when the UAV camera is no longer vertically downward. In contrast, the proposed method, which recovers elevation information of feature points from the satellite map, demonstrates robustness to camera pitch variations, making it more suitable for real-world UAV visual geolocalization tasks.

F. Runtime Analysis and Ablation Study

We further report the average per-frame runtime of each module on both the datasets, as summarized in Table VI. On the SEULoc dataset, the overall processing time is 0.263 s per frame on average, while on the RealUAV dataset the total runtime is reduced to 0.135 s per frame. These results demonstrate that the proposed method achieves practical computational efficiency for UAV visual geolocalization under GNSS-denied conditions, indicating that the system is feasible for near-real-time deployment in IoT scenarios.

In addition, an ablation study was conducted to evaluate the contributions of the elevation recovery module and the joint pose optimization module in the proposed method. As shown in Table VII, removing the elevation estimation module (Ablation Study 1) leads to a notable degradation in both mean and RMSE on SEULoc02 and SEULoc03. Although Ablation Study 1 achieves slightly better results than our full method on the three RealUAV scenarios and SEULoc01, this is because, when the UAV camera is set to a nadir

TABLE VII
RESULTS OF ABLATION STUDY ACROSS DIFFERENT SCENES

Scene	Ablation Study 1		Ablation Study 2		Whole	
	Mean(m)	RMSE(m)	Mean(m)	RMSE(m)	Mean(m)	RMSE(m)
city1	6.07	6.95	7.65	10.02	6.21	7.48
city2	6.14	7.22	8.48	11.01	6.36	7.80
village1	5.83	6.02	6.99	9.50	5.92	6.17
SEULoc01	1.99	2.26	3.60	4.56	2.48	3.14
SEULoc02	10.37	10.89	3.68	4.44	2.84	3.21
SEULoc03	29.05	27.13	4.58	7.75	3.68	3.93

Ablation Study 1: Removing the elevation recovery module and estimating coarse geographic positions using homography only.

Ablation Study 2: Removing the joint optimization module.

TABLE VIII
IMPACT OF ADDED ORIENTATION NOISE ON LOCALIZATION ACCURACY

Noise std	village1		SEULoc01	
	Mean error (m)	Change	Mean error (m)	Change
1	6.43	+0.51	2.58	+0.10
3	7.33	+1.41	3.81	+1.33
5	7.65	+1.73	4.94	+2.46
7	8.04	+2.12	6.03	+3.55
9	8.56	+2.64	6.59	+4.11

view (-90°), the homography-based geolocation method used in Ablation Study 1 becomes more advantageous, whereas our elevation estimation inevitably introduces certain errors. However, as the camera tilt angle changes—which is also more representative of practical UAV applications—our full method significantly outperforms Ablation Study 1. Ablation Study 2 removes the joint optimization module and directly estimates the UAV pose using feature points on the satellite image after elevation recovery. The results show that the full method consistently outperforms Ablation Study 2 across all the datasets, further demonstrating the effectiveness of the joint optimization module in improving localization accuracy.

G. Localization Accuracy Under Different Levels of Geographic Position Noise and Weighting Parameters

To assess the robustness of the proposed pose-fusion backend to inaccuracies in geographic pose estimation, we inject zero-mean Gaussian noise with different Stds into the geographic observations and perform full-trajectory optimization. As shown in Table VIII, the mean localization error increases gradually and smoothly as the noise level rises, demonstrating that the system remains stable under a wide range of perturbations.

Both Village1 and SEULoc01 exhibit a consistent upward trend with increasing noise, but with different slopes due to the distinct scene characteristics and feature distributions of the two datasets. Importantly, neither dataset shows abrupt error spikes or divergence, indicating that the fusion framework effectively suppresses the influence of corrupted geographic measurements. This robustness is largely attributed to the combined effects of the Huber kernel and the adaptive geographic weighting factor α_i , which automatically attenuate unreliable geographic constraints while preserving useful global cues.

In addition, to investigate how the weighting of the geographic pose constraint influences the stability and accuracy

TABLE IX
LOCALIZATION ACCURACY UNDER DIFFERENT VALUES OF THE INFORMATION MATRIX WEIGHT β

β	village1		SEULoc01	
	Mean error (m)	RMSE (m)	Mean error (m)	RMSE (m)
0.5	5.98	6.19	2.59	3.31
1.0 (ours)	5.92	6.17	2.48	3.14
2.0	6.15	6.37	2.71	2.89
5.0	6.66	6.82	3.13	3.42
10.0	7.03	7.29	3.68	3.96

of the proposed SLAM-Geo fusion framework, we perform a sensitivity analysis by scaling the geofactor information matrix with a coefficient β . For each β setting, the system is independently executed ten times, and all the reported values are the averaged results to reduce randomness and ensure statistical reliability. Table IX summarizes the performance on two representative sequences (Village1 and SEULoc01). The results reveal several consistent observations.

First, when β remains within a moderate range (0.5–2.0), the system maintains stable localization performance, with only slight fluctuations in mean error and RMSE. This demonstrates that the proposed optimization framework is not overly sensitive to the precise value of β , indicating that the fusion between visual constraints and geographic priors is well-balanced and inherently robust. Second, when β is significantly increased (e.g., $\beta = 5.0$ or 10.0), the localization accuracy gradually decreases on both the datasets. This trend is expected: excessively strong geographic constraints start to dominate the optimization, forcing the system to overfit the noisy or biased geographic observations, thereby weakening the contribution of the more reliable relative visual constraints. Importantly, the degradation is smooth and monotonic rather than abrupt, showing that the optimization process remains stable even under aggressive weighting.

H. Comparison to Other Studies

Due to the lack of publicly available source code and datasets, we summarize several recent representative studies on UAV scene-matching positioning approaches, as shown in Table X. SAVL [35] designs a feature extraction strategy that adapts to different scenes for UAV visual geolocation. GLVL [15] implements a framework that combines large-scale retrieval and fine-grained matching into end-to-end joint training. Qiu et al. [24] design a spatial resolution adaptive (SRA) module and an image moment extraction (IME) module. The former ensures that the online processed aerial images have approximately equal spatial resolutions to the preprocessed satellite images. However, the above methods rely on 2-D satellite images, which requires UAVs to capture strictly downward-looking views. In contrast, our method recovers the elevation of satellite imagery, enabling accurate localization even under oblique-view UAV perspectives. Yao et al. [17] use DEM to achieve perspective-robust UAV localization; nevertheless, the low resolution of DEM data limits its applicability to low-altitude urban environments. Xu et al. [13] integrate UAV geo-pose estimates with SLAM to

TABLE X
COMPARISON OF REPRESENTATIVE UAV GEOLOCALIZATION METHODS

Research	Year	Method	Dataset	Known Initial	Experiments	Position Error
Yao et al. [17]	2024	3D–2D matching (DEM) + SLAM	Self-collected	Yes	Flight height: 200–1167 m; 2D map + DEM	6.7 / 8.1 / 6.9 m
SAVL [35]	2025	2D matching	RealUAV	Yes	2D map; vertically downward camera	8.71 m
GLVL [15]	2023	Joint large-scale retrieval and fine-grained matching	VTRN + RSSDIVCS	No	2D map; vertically downward camera	2.39 m
Ye et al. [25]	2024	Coarse-to-fine	University-1652; Self-collected	No	Oblique-view; geo-tagged orthographic references; Flight altitude: 150 / 4000 m	0.1 / 10 m
Xu et al. [13]	2024	Image matching + SLAM–absolute pose fusion	Self-collected + Simulation	Yes	2D map; Flight altitude: 500 m	12.47 / 7.53 / 10.32 m
Qiu et al. [24]	2025	2D matching + SRA + IME	Self-collected	Yes	2D map; vertically downward camera; Flight altitude: 500 m	22.33 / 20.08 m
Ours	–	Retrieval + Elevation Recovery + SLAM	RealUAV + Self-collected	No	Oblique-view; 2D map; Flight altitude: 105–200 m	6.21 / 6.36 m, 2.48 / 2.84 / 2.88 m

obtain smooth trajectory estimation. However, such methods require the UAV’s initial location to be known, which is restrictive in practical deployments. Ye et al. [25] adopt a coarse-to-fine strategy consisting of retrieval followed by matching; even so, direct retrieval is computationally inefficient. Our DCMU module performs HMM-based image retrieval, which improves retrieval accuracy compared with direct retrieval approaches.

V. CONCLUSION

This article addresses the challenge of UAV localization in GNSS-denied environments within IoT-enabled intelligent environments and proposes a novel visual geolocalization approach. The proposed method uses a spatiotemporally constrained candidate map retrieval strategy to adaptively constrain the satellite image search space based on real-time UAV poses, enabling efficient and accurate candidate map filtering. In addition, it introduces an elevation recovery mechanism that operates independently of DEM or DSM, by leveraging covisibility across adjacent UAV frames and their feature correspondences with satellite imagery to autonomously extract elevation information. This design effectively addresses localization failures that arise under oblique viewing angles. Finally, a factor graph joint optimization framework is developed to collaboratively refine both the UAV’s absolute geographic pose and its local relative trajectory, significantly improving the smoothness and stability of localization while mitigating the pose discontinuities typically observed in traditional single-frame-based approaches. Extensive experiments conducted on the RealUAV and SEULoc datasets demonstrate the superior performance of the proposed method in terms of both accuracy and stability. Compared with established baselines such as ORB-SLAM3 and ElevPnP, the method notably reduces positioning errors and suppresses pose instability in complex low-altitude urban environments.

Specifically, during transitions from vertical to oblique camera views, the method achieves an 89.5% improvement in positioning accuracy over conventional approaches, demonstrating its effectiveness in enabling collaborative operations of IoT agents in GNSS-denied conditions.

To meet the demands of future UAV and multiagent collaboration in IoT applications, future work will explore an edge collaborative optimization framework. By integrating federated learning with distributed factor graphs, this framework aims to facilitate the sharing of positioning information among UAV clusters across edge nodes, thereby enhancing positioning efficiency and scalability in large-scale collaborative scenarios. In addition, inertial sensors are also being considered for integration into our positioning system to further enhance its robustness.

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